

Naive Bayes Foundations: A Bayes' Theorem Problem

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note 82](#)).
- **Leads to:** Naive Bayes ([Note 87](#)); MAP estimation ([Note 633](#)); Gaussian mixture model (GMM) ([Note 640](#)).

1. Overview

Key point: Three machines make markers; a random marker turns out defective. Bayes' theorem, with the total probability rule for the evidence, gives the chance it came from each machine.

This Note works through a classic problem with Bayes' theorem. The Note also introduces the **law of total probability** (G-1053), the usual way to compute the evidence $P(B)$ in the denominator. Naive Bayes computes its evidence the same way.

2. The problem

Key point: Production shares 20%, 30%, 50%; defect rates 5%, 3%, 1%.

A factory makes markers on three machines:

Machine	Share of all markers	Defect rate of that machine
M1	20%	5%
M2	30%	3%
M3	50%	1%

A marker is picked at random from a batch and turns out to be **defective**. What is the probability that it was made by M3?

Figure 1 shows the idea before any formula. In a typical batch of 1000 markers, M1, M2 and M3 make 200, 300 and 500. Their defect rates leave 10, 9 and 5 defective markers. Knowing the picked marker is defective means it is one of these 24, and only 5 of the 24 came from M3.

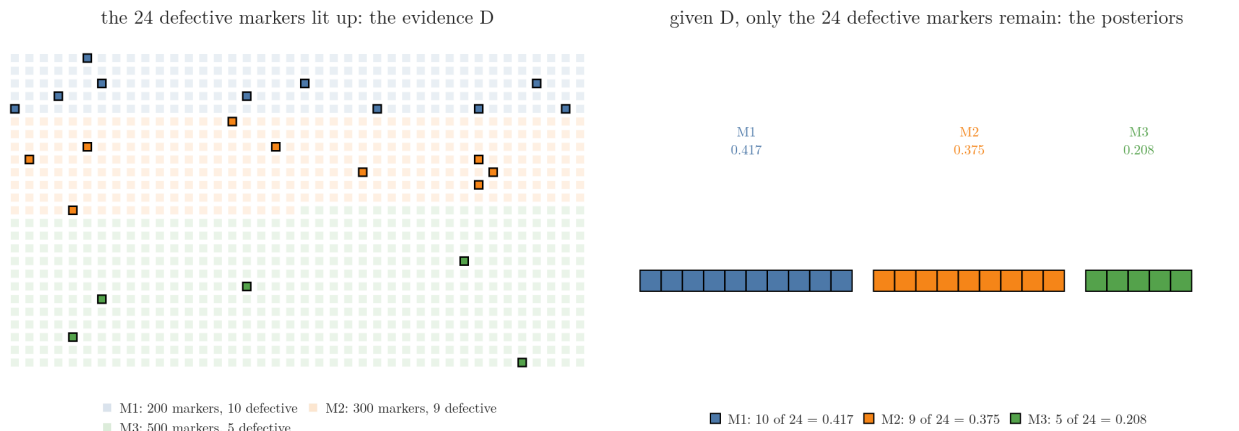


Figure 1: A batch of 1000 markers, one square each, coloured by machine. The 24 defective ones light up (10 from M1, 9 from M2, 5 from M3); given that the marker is defective, only those 24 remain, and M3 supplies 5 of 24 = 0.208.

3. Writing down what we know

Key point: The production shares are priors $P(M)$; the defect rates are likelihoods $P(D | M)$.

Let D be “the marker is defective”. In probability language:

- **Priors** (G-1565): $P(M1) = 0.2$, $P(M2) = 0.3$, $P(M3) = 0.5$.
- **Likelihoods** (G-1086): $P(D | M1) = 0.05$, $P(D | M2) = 0.03$, $P(D | M3) = 0.01$.

The question asks for $P(M3 | D)$. By **Bayes’ theorem** (G-269),

$$P(M3 | D) = \frac{P(D | M3) \times P(M3)}{P(D)}$$

Everything on the right is known except $P(D)$, the overall probability that a marker is defective.

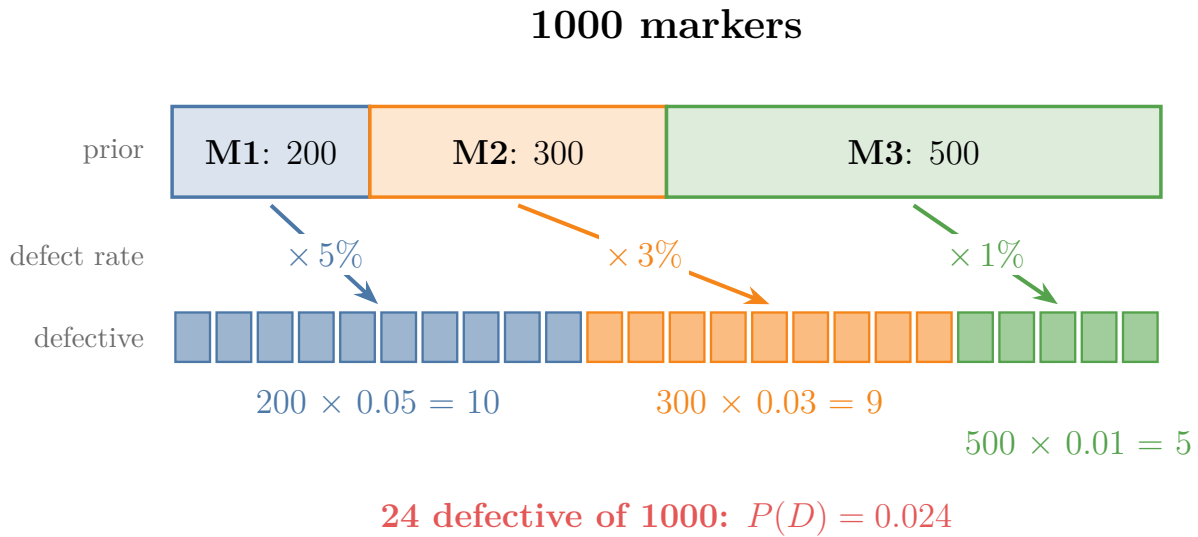


Figure 2: The problem as counts in a batch of 1000 markers. Top: the priors, with widths to scale. Bottom: one square per defective marker, from each machine's count times its defect rate

Figure 2 turns the priors and likelihoods into expected counts for 1000 markers. Watch the bottom row: M3 makes the most markers but supplies the fewest defective squares.

4. The evidence: total probability

Key point: A defective marker came from exactly one machine, so $P(D)$ is the sum of the three “defective and from machine i ” probabilities.

A defective marker must have come from M1, M2 or M3, and from only one of them. These three cases are mutually exclusive (the mutually exclusive events Note), so their probabilities add:

$$P(D) = P(D \cap M1) + P(D \cap M2) + P(D \cap M3)$$

Each term is a likelihood times a prior, by $P(A \cap B) = P(B | A)P(A)$ from the Bayes' theorem proof:

$$P(D) = P(D | M1)P(M1) + P(D | M2)P(M2) + P(D | M3)P(M3)$$

This sum is the law of total probability. With numbers:

$$P(D) = 0.05 \times 0.2 + 0.03 \times 0.3 + 0.01 \times 0.5 = 0.010 + 0.009 + 0.005 = 0.024$$

So 2.4% of all markers are defective. Figure 3 shows the same calculation as a **probability tree** (G-1573).

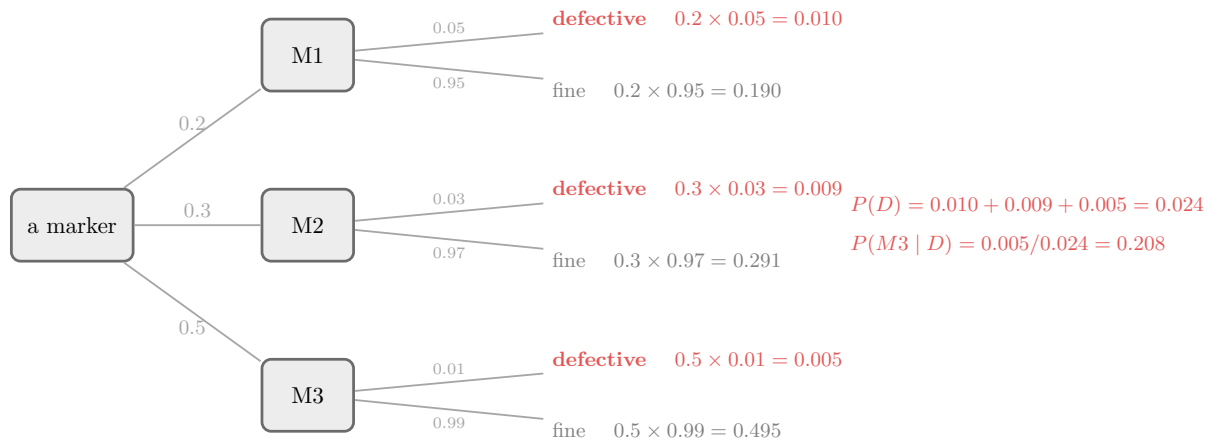


Figure 3: Probability tree: each path multiplies its branches

5. The answer

Key point: $P(M3 | D) = 0.005 / 0.024 = 0.208$. M3 makes half the markers but only about a fifth of the defective ones.

$$P(M3 | D) = \frac{0.01 \times 0.5}{0.024} = \frac{0.005}{0.024} = 0.208$$

The same calculation for every machine:

Machine	Prior $P(M)$	Joint probability (G-986) $P(D \cap M)$	Posterior (G-1536) $P(M D)$
M1	0.20	0.010	0.417
M2	0.30	0.009	0.375
M3	0.50	0.005	0.208

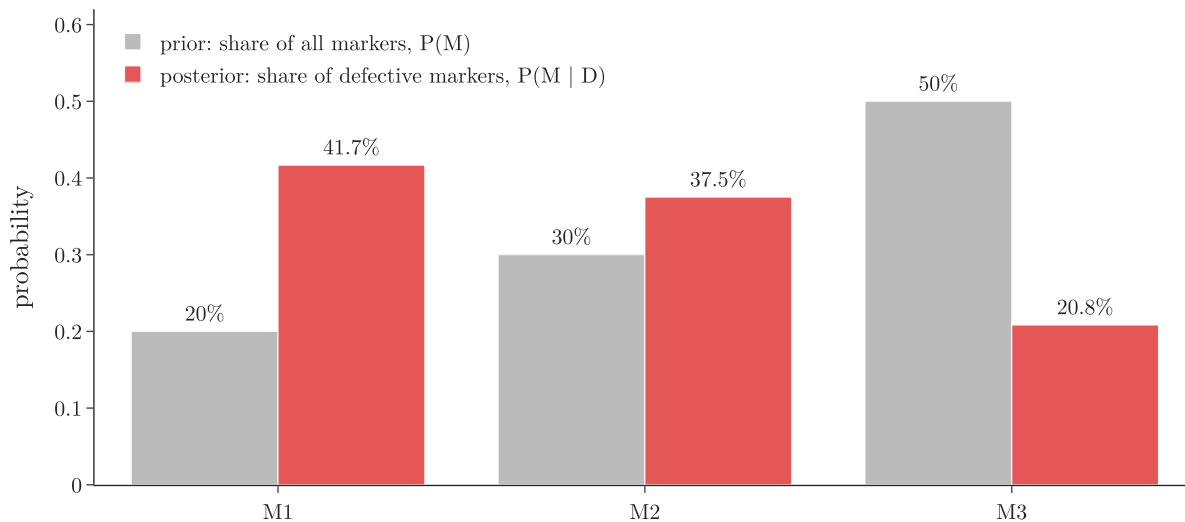


Figure 4: Share of all markers against share of defective markers

Seeing that the marker is defective changes the picture completely (Figure 4). Before, M3 was the most likely source (50%). After, M3 is the least likely (21%), because M3 rarely makes defects.

An everyday picture: three cooks share a kitchen, and one dish comes out burnt. The cook who makes the most dishes is not the likely culprit if that cook almost never burns anything. M1 makes only 20% of markers but 42% of the defective ones.

The three posteriors add up to 1, as they must: the marker came from some machine. A simulation of a million markers gives $P(M3 | D) = 0.2095$, matching 0.208.

Python: The whole calculation.

```
prior = {"M1": 0.2, "M2": 0.3, "M3": 0.5}
p_defect = {"M1": 0.05, "M2": 0.03, "M3": 0.01}

joint = {m: prior[m] * p_defect[m] for m in prior}
p_D = sum(joint.values()) # 0.024
{m: joint[m] / p_D for m in prior} # 0.417, 0.375, 0.208
```

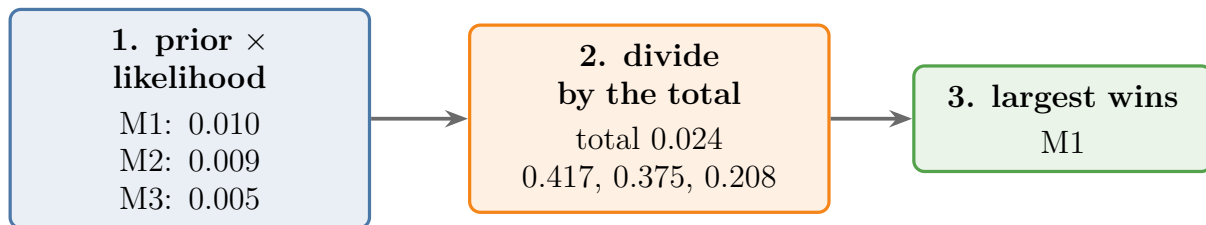
6. The pattern for Naive Bayes

Key point: For each class: prior $\times$ likelihood. Divide by their total. The largest posterior wins.

A Naive Bayes classifier follows exactly this procedure. The machines become the classes, the values of the **target** (G-1949; the output we predict). “Defective” becomes the observed **features** (G-772; the input variables, one column each of the data table):

1. for every class, multiply its prior by the likelihood of the observed features;
2. divide each product by their sum (the evidence) to get posteriors that add to 1;
3. predict the class with the largest posterior.

Since the evidence is the same for every class, step 2 does not change which class wins; it only turns the scores into probabilities.



step 2 rescales every score by the same number: the winner does not change

Figure 5: The three Naive Bayes steps on the marker problem

Figure 5 runs the three steps on the marker numbers: M1 has the largest score both before and after dividing by 0.024.

7. Summary

- Priors: production shares; likelihoods: defect rates; posterior: which machine, given a defect.
- Law of total probability: $P(D) = \sum_i P(D | M_i)P(M_i)$.
- Answer: $P(M3 | D) = 0.208$; M1 is the most likely source (0.417).

8. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 5 | Problem based upon Bayes Theorem”, YouTube, <https://www.youtube.com/watch?v=aAEHjXDHtbE>

9. Key terms

Term	Meaning
Law of total probability	$P(B) = \sum_i P(B A_i)P(A_i)$, when the A_i are mutually exclusive and cover every case
Probability tree	A diagram in which each path multiplies the probabilities along its branches
Joint probability	The probability that two events happen together, $P(A \cap B)$